

1. Introduction

1.1 Background

CSA engineers need fast and reliable access to technical information. Today this information sits in many places: file systems, document tools, and separate team folders. This slows engineering effort down. CRAFT is built to fix this. CRAFT is a working tool for real engineering tasks. It is not a research experiment.

CRAFT combines a Large Language Model (LLM — an artificial intelligence model trained on text, which can read and write natural language) with a shared knowledge base. The knowledge base holds CSA documents, standards, and past mission data. CRAFT uses a software agent: a program driven by the LLM that can plan a task, look up information, write an answer, and record what it did.

1.2 Purpose

CRAFT gives CSA engineers a secure and traceable tool. It answers technical questions, helps estimate cost, finds risks, monitors for unusual system behaviour, and writes reports. This document lists what CRAFT must do, from the point of view of the people who will use it.

1.3 Scope

This document covers CRAFT as a production system used across CSA engineering teams. It covers document handling, document search, risk review, cost estimation, anomaly detection, mission planning support, and report writing. It also covers connections to CSA's existing systems.

CRAFT uses an Agentic-RAG design. RAG means Retrieval-Augmented Generation: the model looks up real documents before it writes an answer. Agentic means a software agent plans and carries out the work, including choosing how to search within the bounds set by UMR-094.

Explicitly out of scope. The following are not part of CRAFT, in this phase or as currently specified:

- Multiple agents by default. CRAFT uses one accountable primary agent. Delegation is the exception, justified case by case under UMR-011.
- Fixed agent role hierarchies. Roles are not imposed in advance. Control is held at the tool boundary under UMR-012.
- Self-evolving agent harnesses. An agent shall not modify its own instructions, tools or code, and shall not modify procedural memory outside the approved change process.
- Free-form shell or command-line access by agents. File search is available only through the parameterised interface in UMR-094.

- Personal devices, individual OneDrive accounts, and production CSA repositories, until each is onboarded through the approved process in UMR-051.
- French-language output and interface, until phase 2.

1.4 Who can change this document

Only CSA's document control process can approve a change. Anyone proposing a change sends it through that process. Approved changes appear in the next issued version.

1.5 Word meanings and precedence

- SHALL — a firm rule. It must be built and it must be verified.
- SHALL NOT — a firm prohibition.
- SHOULD — recommended, but not required.
- MAY — permitted and optional. MAY is never used to express a prohibition in this document.
- WILL — a statement of what is planned or expected.
- CONFIGURABLE — a value an authorized person can change without changing this document. Every configurable value has a parameter identifier and an entry in Annex B.
- BASELINE VALUE — the value approved for use now. An interim baseline value applies until test evidence replaces it.

The Type column has these values: Mandatory, Mandatory (revised), Mandatory (new), Goal, and Withdrawn. A Withdrawn identifier is retained as a pointer so traceability does not break. No mandatory requirement may depend on a Goal.

Precedence. Section 4.1 holds the governing requirement text. Section 4.2 gives additional detail for the software agents. Where Section 4.2 and Section 4.1 appear to differ, Section 4.1 governs. Where a Rationale appears to differ from its requirement, the requirement text governs.

1.6 Requirement tag meanings

Tag	Stands for	Used for
UMR	User and Mission Requirement	The governing requirements in Section 4.1
AAR	Agent Architecture Requirement	How agents are built and bounded (Section 4.2.1)
ARR	Agent Reasoning Requirement	How agents plan, search and answer (Section 4.2.2)
HITL	Human-In-The-Loop Requirement	Human review and approval (Section 4.2.3)
ASG	Agent Safety and	Safety, limits and oversight

Tag	Stands for	Used for
	Governance Requirement	(Section 4.2.4)
UC	Use case	Appendix B of the source requirements document, as corrected in Annex C
PARAM	Configurable parameter	Annex B register entries
OP	Open point	Annex E items awaiting a CSA decision
RC	Review comment	Annex F reconciliation register
ADR	Architecture Decision Record	Held in the Architecture Design Document

1.7 Verification methods

- I — Inspection. Read the product, the configuration, or the document.
- A — Analysis. Calculate or reason from data.
- D — Demonstration. Run the system and observe the behaviour.
- T — Test. Run a defined procedure and compare results against a stated criterion.

1.8 Terms used in this document

Term	Plain meaning
Agent	A program, driven by the language model, that plans and carries out a task
API	Application Programming Interface — a defined way for one system to call another
Azure	Microsoft Azure — the cloud platform CRAFT runs on
BM25	A standard keyword ranking method. It scores a document by how often the query words appear in it, adjusted for length and word rarity
Chunk	A piece of a document, stored and searched as a unit
Context window	The total amount of text a language model can consider in one request, measured in tokens
Deduplicate	Remove repeated passages so the same text is not counted twice in the evidence
Delegation	The primary agent asking a subagent to

Term	Plain meaning
	complete a bounded subtask
Dense retrieval	Search by meaning, using embeddings. Also called vector search or semantic search
Embedding	A list of numbers that represents the meaning of a piece of text
Ephemeral	Held only for the life of the session, then destroyed
Episodic memory	Governed conversation and task history for the current session
Fusion	Combining two ranked result lists into one ranked list, after putting their scores on a common scale
Grounding	Writing answers only from retrieved evidence
Knowledge graph	A stored map of facts and the links between them
MATLAB	A commercial engineering calculation and modelling product
Normalize	Put scores from different search methods onto a common scale so they can be compared
Parameterized interface	A tool that accepts a fixed set of named inputs, rather than a free-form command string
PBMM	Protected B, Medium integrity, Medium availability — the Government of Canada cloud security control profile CRAFT is assessed against
Primary agent	The single agent accountable to the user for a session and for the final answer
Procedural memory	Approved guidance for a repeated task, such as instruction files, runbooks and templates
Prompt	The written instruction given to the language model
Prompt injection	Hidden instructions placed inside a document, intended to change agent behavior

Term	Plain meaning
Retrieval path	The record of which search tools an agent used, in what order, to find the evidence behind an answer
REST, GraphQL	Two common styles of API
Recall@k, Precision@k, MRR, nDCG, F1	Standard measures of how good a ranked list of search results is
SAP	The commercial product used as CSA's financial system of record
Semantic memory	Curated, source-backed durable knowledge, held independently of any one conversation
Session	One continuous conversation between a user and CRAFT, from opening to close
Source agreement	Whether several independent sources say the same thing
Sparse retrieval	Search by keyword, such as BM25
SSC	Shared Services Canada
STK	Systems Tool Kit — a commercial mission analysis product
Token	A word-piece of text. Models measure text length in tokens
Unsupported claim	A statement in an answer that no retrieved passage backs up
Z-score	A measure of how far a data point sits from the normal range

2. Related Documents

2.1 Parent documents

No parent requirements document exists above CSA-CRAFT-RD-0001. CRAFT is the top of its own requirement tree. AD-01, CSA-SE-PR-0001, remains the process authority for how this document is written and controlled.

Document number. CSA has confirmed the document number as CSA-CRAFT-RD-0001. The source file header carried CSA-SEFM-RD-001, which is superseded. The header shall be corrected at reissue.

2.2 Documents CRAFT must follow

ID	Reference	Title
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ID	Reference	Title
AD-01	CSA-SE-PR-0001	Systems Engineering Methods and Practices
AD-02	Canada.ca	Responsible use of artificial intelligence in government
AD-03	NASA/SP-2016-6105 Rev 2	NASA Systems Engineering Handbook
AD-04	ECSS-E-ST-10C	European Cooperation for Space Standardization Systems Engineering Standard
AD-05	CSA-SEC-POL-003	CSA Data Classification Policy
AD-07	Canada.ca	AI governance principles
AD-08	CSA-ST-GDL-0001	Technology Readiness and Risk Assessment Guidelines

2.3 Documents used for background

ID	Reference	Title	Status
RD-01	CSA-CRAFT-AD-0001	CRAFT Architecture Design Document	Draft A1 is a cover page only. Treated as not available. See Section 5.3
RD-02	N/A	CRAFT Intern Proposal	Available
RD-03	CSA-SE-GDL-0001	Systems Engineering Mission Tailoring Guidelines	Available
RD-04	CSA-SEFM-RD-0001, Appendix B	CRAFT Operational Use Cases	Recovered in P9. UC-A1, UC-B1, UC-C1 to UC-C9
RD-05	CSA-SEFM-RD-0001, Appendix C	CRAFT Traceability Tables	Recovered in P9. Completed in P11
RD-06	tbs-sct.canada.ca	Treasury Board Directive on Automated Decision-Making	CRAFT does not claim compliance. See OP-14
RD-07	CSA-SEFM-RD-0001, Appendix A	User Roles, Operational Assumptions, Risk	Controlled copy. The infrastructure estimate is

ID	Reference	Title	Status
		and Initial Infrastructure Estimate	withdrawn in full
RD-08	requirements-analysis.docx	CRAFT Requirements Technical Review Commentary and Responses	Reconciled in Annex F, RC-01 to RC-19

2.4 Approval block

Role	Authority
Prepared by	Senior Systems Engineer, Sun-Earth System Science
Reviewed by	Senior Systems Engineer, CSA Finance
Reviewed by	Senior Engineer, Electrical, Space Science and Technology
Recommended by	Senior Manager, Sun-Earth System Science
Recommended by	Engineering Manager, Space Science and Technology
Recommended by	CSA Finance
Approved by	Director, Sun-Earth System Science

If a further approval authority is required for the PBMM authorization, it shall be added by name and title through the change process.

3. What CRAFT Is Meant to Do

3.1 Main goals

CRAFT gives CSA a safe and traceable place where authorized people can find, check, and use technical knowledge and engineering data. People work with CRAFT through a conversation, asking questions in plain language and following up. CRAFT helps engineers make better decisions.

3.2 How CRAFT works, step by step

In the proof of concept, CRAFT imports documents from the controlled Engineering Costing corpus (UMR-051, UMR-091). It checks the classification of each document against the Protected B ceiling. It splits each document into chunks, converts each chunk into an embedding, and keeps a search index and a knowledge graph.

A user opens a session. One primary agent is accountable for that session (UMR-095). The user asks a question. CRAFT plans the task. It searches using the default hybrid method: keyword and meaning-based retrieval, normalised onto a common scale, fused, and deduplicated. Where the question needs a different approach, the primary agent may select another authorized retrieval tool and iterate (UMR-094). Retrieval stays selective: CRAFT places the relevant evidence in the model request, not as much as will fit.

CRAFT writes an answer from the retrieved evidence, records which retrieval path it used, and scores its confidence from evidence coverage, citations, traceability, source agreement, retrieval quality, and any claims it could not support. If the task is high-risk, or the confidence is below the approved threshold, a person reviews the result before it is used.

3.3 Who uses CRAFT

User group	Status	What they do	Notes
Engineers and analysts	Built now	Ask questions, run analysis, draft reports	Hold the Requester role, and Reviewer where appropriate
Technical authorities and delegated authorities	Built now	Approve recommendations, high-risk actions and final reports	Hold the Approver or Backup approver role
Governance, audit and security staff	Built now	Review the audit record, traceability reports and PBMM control evidence	Hold the Auditor or Security authority role
Platform support staff	Built now	Run and maintain the platform	Hold the Administrator role. Cannot approve requests or alter audit records
Executives	Planned for later	Read summaries and approved results	Not built yet

3.4 Roles and responsibilities

Required by UMR-065. Every CRAFT user holds at least one role. A role grants only the permissions listed for it. These are user roles. Agents do not hold user roles; agent permissions are bound at the tool level under UMR-012.

Role	Who it is for	May do	Shall not do
Requester	Any authorized CSA	Open sessions, ask	Approve their own

Role	Who it is for	May do	Shall not do
	engineer or analyst	questions, run searches, request cost estimates and risk reviews, run mission planning comparisons, generate draft reports, use the session workspace, see their own history	request. Mark a report final. Change any configurable parameter. Read the audit store
Reviewer	Subject-matter experts and technical leads	Everything a Requester may do. Review low-confidence answers routed under UMR-008. Correct or reject an agent recommendation	Give final approval for a high-risk action. Change any configurable parameter
Approver	Technical authorities, project managers, delegated financial or risk authorities	Approve, reject or edit any recommendation before it becomes final. Approve high-risk actions. Mark a report final. Approve changes to the official risk list	Approve a request they submitted themselves (UMR-065.1). Change or delete an audit record
Backup approver	An alternate for each Approver. Named holders TBC (PARAM-37)	Act as Approver when a request escalates on timeout under UMR-025	As for Approver
Parameter owner	The CRAFT technical authority and named delegates	Change a configurable parameter in Annex B, with a recorded reason and evidence. Approve the authorized tool set under UMR-012 and UMR-094	Approve operational requests. Change an audit record

Role	Who it is for	May do	Shall not do
Auditor	Governance reviewers, internal audit, security assessors	Read and export the full audit record, including retrieval paths, delegation records and code execution records. Produce traceability reports	Hold the Administrator role. Submit, review or approve operational requests
Administrator	CRAFT platform support staff	Manage the platform, deploy releases, manage accounts and role assignments, manage connectors and the tool registry	Change or delete an audit record, prevented technically by ASG-008. Approve operational requests. Hold the Auditor role
Security authority	The CRAFT security assessor under the PBMM authorization	Review the PBMM control mapping, classification rejections, the unlabelled-document population, code execution logs, blocked tool attempts, and workspace exports	Change the configuration or approve operational requests

Least privilege. A user shall be given the lowest role that lets them do their work. Role assignments shall be reviewed at the approved interval (PARAM-36).

Phase note. The proof of concept may operate with a small number of people covering several roles, but the separation in UMR-065.1 shall hold: the requester and the approver shall be different people, and the auditor shall not be the administrator.

4. What CRAFT Must Do

4.1 Core requirements

Column meanings: ID is the tracking number. Type is how firm the rule is. Requirement is the rule. Rationale says why it exists. V is the verification method (I, A, D, T). Note records what changed.

ID	Type	Requirement	Rationale	V	Note
UMR-001	Mandatory (revised)	CRAFT shall accept questions written as free text, in English or in French, within a conversational session as defined in UMR-095. CRAFT shall detect the language of the question. Language detection accuracy shall be measured against the approved acceptance question set. The minimum accuracy is a configurable setting (PARAM-01).	Engineers must be able to ask in their own words and follow up.	T	Linked to the session model in V3
UMR-002	Mandatory (revised)	CRAFT shall retrieve documents that match the question, and shall attach to every retrieved passage the identity of	The answer must be checkable.	D	Citation content in UMR-006

ID	Type	Requirement	Rationale	V	Note
UMR-003	Mandatory (revised)	the document it came from. CRAFT shall supply the language model with enough retrieved evidence to answer the questions in the approved acceptance set. The minimum total request context shall be a configurable setting (PARAM-02), covering instructions, user query, conversation history, retrieved evidence and generated output. It is not the size of a stored chunk; chunk size is PARAM-39. Models with a larger context window are permitted.	The earlier wording did not say whether 8192 meant a chunk or the model context window. It is the request context, and it is a compatibility floor rather than a target. A larger window is not a reason to send more evidence: long prompts dilute the passages that matter, which degrades both answer quality and the grounding measure in UMR-010.	T	Selectivity rule added in V3 from RC-03

ID	Type	Requirement	Rationale	V	Note
		Retrieval shall remain selective and relevant. CRAFT shall not fill the available context indiscriminately.			
UMR-004	Mandatory (revised)	CRAFT shall find and rank matching documents using both keyword retrieval and meaning-based retrieval. CRAFT shall normalise the scores from each method onto a common scale, fuse them into a single ranking, and deduplicate the result. CRAFT shall not simply append the top results of each method. The number of results, the filter threshold,	The review asked how results from the two methods are ranked against each other. Appending raw lists distorts what the model sees, because two methods produce scores on different scales. A threshold is acceptable once measured; the earlier 0.72 was unacceptable only because it was asserted without evidence.	T	Normalise, fuse, deduplicate added in V3 from RC-04. Top-5-append fallback withdrawn. 0.72 restored as a candidate to validate. Measurement scope widened to UMR-094

ID	Type	Requirement	Rationale	V	Note
		the fusion method and the ranking method shall be configurable settings (PARAM-03). Threshold values, including the candidate value of 0.72, shall be validated against measured retrieval performance on the approved acceptance set before use in operation. Quality shall be measured across all retrieval paths, including agent-selected retrieval under UMR-094, not only the default path.			
UMR-005	Mandatory (revised)	CRAFT shall retrieve using BM25 keyword	Fixing the weights for Phase 1 accelerates	I, T	Phase 1 weights fixed, interface

ID	Type	Requirement	Rationale	V	Note
		search and vector search together, behind a single common retrieval interface. The interface shall allow the retrieval methods, the fusion weights, and any reranking step to be changed without changing the calling code. The weights shall be a configurable setting (PARAM-04). Phase 1 baseline: 0.7 semantic and 0.3 lexical.	delivery. A common interface means later phases can add configurable weights, reranking, metadata search and other algorithms without refactoring.		retained, RC-06
UMR-006	Mandatory (revised)	Every answer shall name, for each fact used: the source document, its version, and the page or section of the cited	People must be able to check the work.	D	"If possible" removed. UMR-060 merged in

ID	Type	Requirement	Rationale	V	Note
		passage. Where the source format has no page or section markers, CRAFT shall give the passage locator it holds.			
UMR-007	Mandatory (revised)	CRAFT shall give every answer a confidence score between 0 and 1. The score shall be computed from: evidence coverage; the presence and quality of citations; source traceability; agreement between independent sources; retrieval quality; and the identification of claims the retrieved evidence does not	Multiplied token probabilities fall towards zero as an answer lengthens, regardless of correctness, and hosted models often do not return them at all. The six named inputs are observable, model-independent and testable.	T	Inputs extended in V3 from RC-07 and RC-08. Removes a hidden constraint on UMR-081

ID	Type	Requirement	Rationale	V	Note
		support. CRAFT shall not compute confidence from the joint probability of generated tokens. CRAFT shall demonstrate that the score separates correct answers from incorrect answers on the approved acceptance set.			
UMR-008	Mandatory (revised)	If an answer's confidence score is below the approved threshold, CRAFT shall route the answer to the assigned human reviewer role before it is used. The threshold shall be a configurable setting per risk tier	A single fixed number for every task contradicts the risk-tiering rule in UMR-010.	T	Baseline value

ID	Type	Requirement	Rationale	V	Note
UMR-009	Mandatory (revised)	(PARAM-06). Interim baseline value: 0.6. If a retrieval result does not meet the configured sufficiency condition, CRAFT shall search again with adjusted terms or a different authorized retrieval tool under UMR-094. The number of retries shall be a configurable setting (PARAM-05).	An uncapped retry conflicts with the loop control in UMR-019.	T	Linked to agent retrieval in P13
UMR-010	Mandatory (revised)	CRAFT shall write answers only from retrieved evidence. CRAFT shall measure grounding against the approved acceptance set, using the approved checking method. The	A number without an agreed method, data set and tool means nothing. Removing the gate entirely leaves no control.	T	Split. Insufficient-evidence behaviour in UMR-086

ID	Type	Requirement	Rationale	V	Note
		pass score, the sample method and the checking tool shall be written down before acceptance testing. The pass score shall be a configurable setting per risk tier (PARAM-07).			
UMR-011	Mandatory (revised)	CRAFT shall use one accountable primary agent for each session. CRAFT shall not impose fixed agent roles and shall not use multiple agents by default. The primary agent may delegate a bounded subtask to a subagent only where delegation demonstrably improves quality, speed,	The review said multi-agent roles were not well defined and it was not clear why they were needed. The response settles it: most tasks need one agent, so the question of how to govern many agents was solving a problem CRAFT does not have. Requiring a recorded justification prevents delegation	I, T	Rewritten in V3 from RC-15 and RC-16. Registered-role-per-agent rule withdrawn. OP-30 closed

ID	Type	Requirement	Rationale	V	Note
		isolation or capability, measured against the criteria in PARAM-45. Every delegation shall record the subtask, the reason, the bounds set, and the result. The primary agent remains accountable for the final response.	becoming the default by habit.		
UMR-012	Mandatory (revised)	Every agent, primary or delegated, shall use only the tools listed in the approved tool registry for its task, including retrieval tools. Tool permissions shall be held under version control and approved by the Parameter owner role.	With fixed roles withdrawn under UMR-011, the tool boundary becomes the primary control surface. What an agent may do is defined by what it may call.	T	Elevated in P13. Now the main agent control, and explicitly covers retrieval tools

ID	Type	Requirement	Rationale	V	Note
UMR-013	Mandatory (revised)	Agents shall run within a budget covering time used, tokens used, tool calls made, reasoning steps taken, delegated subtasks, and estimated cost. Each limit shall be a configurable setting per task type, per requester and per risk tier (PARAM-09 to PARAM-13). When a limit is reached, CRAFT shall stop safely, return any useful partial result, record the reason, and ask the user what to do next. For high-risk tasks, CRAFT shall escalate to	A step count alone does not reflect real cost, time or risk. Delegation adds a cost dimension that did not previously exist.	T	Delegation budget added in P13

ID	Type	Requirement	Rationale	V	Note
UMR-014	Mandatory (revised)	the assigned human role. CRAFT shall hold three kinds of memory. Episodic memory is the governed conversation and task history for a session. Semantic memory is curated, source-backed durable knowledge, held independently of any one conversation; an entry shall cite its source and shall be reviewable. Procedural memory is approved guidance for a repeated task, such as instruction files, runbooks and templates. Procedural memory	The review asked what semantic memory means and proposed episodic memory as conversation history. The response tightens both: semantic memory must be curated and source-backed, not whatever the system happened to retain, and procedural memory must be approved rather than self-written. Uncontrolled self-modification of procedures is how an agent's behaviour drifts without anyone approving	I, T	Definitions tightened in V3 from RC-11 to RC-14

ID	Type	Requirement	Rationale	V	Note
		shall be held under version control and shall be changed only through the approved process; an agent shall not modify it. Each kind shall have a defined retention period (PARAM-27), a defined deletion path, and isolation per user and per data classification.	the change.		
UMR-015	Mandatory	Every message between agents shall include the sending agent identifier, the action, the input, the output, a confidence score and a timestamp.	Needed to control and review agent behaviour.	I	Unchanged
UMR-016	Mandatory (revised)	CRAFT shall allow an agent to be	The review read the original	D	Clarified, RC-09

ID	Type	Requirement	Rationale	V	Note
		started or stopped while the platform continues to serve requests. Requests already running shall not fail. This concerns platform availability during agent lifecycle changes. Session persistence is covered by UMR-095.	wording as implying persistent session-bound agents. The two concerns are now separated.		
UMR-017	Mandatory (revised)	An agent shall not read, change or delete data, system functions or system settings outside the permissions recorded for its tools in the approved tool registry.	Protects data and keeps control of the system.	T	Bound to the tool registry in P13
UMR-018	Mandatory (revised)	CRAFT shall break a complex request into steps and record the plan. Where	Needed for tasks with several steps.	T	Delegation recording added

ID	Type	Requirement	Rationale	V	Note
		a step is delegated, CRAFT shall record which subagent executed it. Verification is against the approved multi-step acceptance tasks.			
UMR-019	Mandatory (revised)	If an agent repeats an action that matches the approved repetition test, and no new information is produced, CRAFT shall stop the task and escalate it to the assigned human role. The number of repetitions allowed shall be a configurable setting (PARAM-14). Interim baseline value: 3.	Prevents wasted and unsafe repeated execution, including repeated retrieval attempts under UMR-094.	T	Aligned with ASG-005
UMR-020	Mandatory (revised)	CRAFT shall filter generated content	A filter with no measured false-block rate can	T	Criteria added

ID	Type	Requirement	Rationale	V	Note
		against the approved content policy list. Filter performance shall be measured against the approved adversarial test set. The minimum block rate and the maximum false-block rate shall be configurable settings (PARAM-08).	make CRAFT unusable.		
UMR-021	Mandatory (revised)	CRAFT shall obtain approval from an authorized person before carrying out any action classified as high-risk under UMR-082.	Protects important engineering decisions from being made by AI alone.	T	One classification scheme
UMR-022	Mandatory (revised)	Every approval request shall show the agent's reasoning, its confidence	A reviewer needs the evidence and the route by which it was found.	D	Retrieval path added in P13

ID	Type	Requirement	Rationale	V	Note
		score, the source documents used, and the retrieval path under ARR-010.			
UMR-023	Mandatory (revised)	A person holding the Approver role defined under UMR-065 shall be able to approve, reject or edit any recommendation before it becomes final.	A person makes the final decision.	D	“Authorized” now defined
UMR-024	Mandatory (revised)	If a recommendation is rejected, CRAFT shall allow the agent to try again using the reviewer’s correction. The number of retries after rejection shall be a configurable setting (PARAM-15).	An uncapped retry loop is a cost and safety risk.	T	Retry cap added
UMR-025	Mandatory (revised)	If no decision is recorded	Silent cancellation	T	Escalation, safety

ID	Type	Requirement	Rationale	V	Note
		within the approval timeout, CRAFT shall escalate the request to the Backup approver. The timeout shall be a configurable setting per risk tier (PARAM-16). Interim baseline value: 24 hours. If no decision is recorded after escalation, CRAFT shall cancel the request and flag it for review. If no Backup approver is named, CRAFT shall not cancel the request; it shall keep it open and flag it. CRAFT shall not cancel a request that reports a high-severity operational	of a safety escalation is not acceptable, and a blank role must not become a silent cancellation.		exception, unnamed-backup case

ID	Type	Requirement	Rationale	V	Note
UMR-026	Mandatory	anomaly. Every approval decision shall be logged with the person's identity, the decision, the time and the reason.	Needed to prove who approved what.	I	Unchanged
UMR-027	Mandatory (revised)	CRAFT shall write every agent action and every approval decision to append-only storage, including retrieval paths, delegation records, workspace exports and code execution records. No platform role, including administrator, shall be able to change or delete a record. Record integrity shall be verifiable by	“Permanent and unchangeable” could not be tested. The new agent capabilities each produce records that must be covered.	T	Scope widened in P13

ID	Type	Requirement	Rationale	V	Note
UMR-028	Mandatory (revised)	the approved cryptographic check. Every version of an agent’s model settings, prompts, tool permissions and procedural memory shall be held under version control, and shall be reviewed and approved by the assigned technical authority before operational use. An agent shall not modify its own settings, prompts, tools, code or procedural memory.	The review confirmed self-evolving agent harnesses are out of scope, and the response extends that to procedural memory.	I	Procedural memory added in V3 from RC-14
UMR-029	Mandatory (revised)	CRAFT shall classify every agent action into one of three groups under UMR-082:	“Reversible where feasible” could not be tested.	T	Split into UMR-087, UMR-088

ID	Type	Requirement	Rationale	V	Note
		read-only, reversible, or irreversible. For a reversible action, CRAFT shall check the action before running it, save the earlier state or a version marker, write a log entry, and provide a way to undo the change.			
UMR-030	Mandatory (revised)	CRAFT shall limit resource use for each user, each session, each task and each agent. The limits shall cover tokens, tasks running at the same time, estimated cost, tool calls, call rate, delegated subtasks, retrieval iterations,	Agent-selected retrieval and delegation are both unbounded without explicit limits: an agent that may choose how to search may also choose to search many times.	T	Retrieval iterations and delegation added in P13

ID	Type	Requirement	Rationale	V	Note
		and code execution time and memory (PARAM-42). Each limit shall be a configurable setting with an interim baseline value in force before first operational use. Soft limits shall warn and reduce the rate of new work. Hard limits shall stop the task safely.			
UMR-031	Mandatory (revised)	CRAFT shall produce mission cost estimates from past mission data. Estimate accuracy shall be measured by a back-test against past missions with known cost. The minimum accuracy	An estimate with no measured accuracy cannot be relied on.	T	Accuracy criterion added

ID	Type	Requirement	Rationale	V	Note
UMR-032	Mandatory (revised)	shall be a configurable setting (PARAM-17). CRAFT shall compare missions using a documented , repeatable comparison method that adjusts for different measurement scales. The method and its weights shall be configurable settings under version control (PARAM-18). Interim baseline method: weighted normalised Euclidean distance.	Locking one formula before testing repeats a defect corrected elsewhere.	I, T	Baseline method
UMR-033	Mandatory (revised)	CRAFT shall find the closest matching past missions to support a new cost estimate. The number	The right count depends on how much history exists.	T	Baseline value

ID	Type	Requirement	Rationale	V	Note
		returned and the similarity cutoff shall be configurable settings (PARAM-19).			
UMR-034	Mandatory (revised)	A cost estimate shall include an uncertainty range and a machine-readable list of the past missions used as evidence.	A reviewer must be able to check the estimate and see how uncertain it is.	D	Defined meaning
UMR-035	Mandatory (revised)	A cost estimate tied to a spending decision or an approved baseline shall be classified as high-risk under UMR-082 and shall require approval before use.	Protects the organization from acting on an unapproved estimate.	T	One classification scheme
UMR-036	Mandatory (revised)	CRAFT shall identify and flag possible risks found in mission documents. The official risk list is	UMR-037 to UMR-040 are mandatory and depend on this capability.	T	Risk standard TBC, OP-08

ID	Type	Requirement	Rationale	V	Note
		handled at Protected B under PBMM. The risk standard CRAFT applies is TBC.			
UMR-037	Mandatory (revised)	CRAFT shall find risks using one or more approved detection methods. The methods used and their settings shall be configurable and under version control.	Locking one method before testing repeats the defect corrected in UMR-005.	I, T	Method set configurable
UMR-038	Mandatory (revised)	Every risk CRAFT reports shall carry a severity level from the severity scale defined in UMR-082.	Helps decide which risks need attention first.	D	One scale
UMR-039	Mandatory	Any change proposed to the official risk list shall require approval by an authorized person	Protects the controlled risk record.	T	Unchanged

ID	Type	Requirement	Rationale	V	Note
UMR-040	Mandatory (revised)	before it is saved. Every risk CRAFT reports shall cite evidence found in the source document, and shall meet the grounding rule in UMR-010.	Prevents invented risks.	T	Linked to grounding test
UMR-041	Mandatory (revised)	CRAFT shall detect and report configured anomalies in approved, monitored data feeds and system health information, using configurable rules, limits and trend checks.	The scope must stay limited to data CRAFT can reach.	T	Report content in UMR-089
UMR-042	Mandatory (revised)	CRAFT shall flag unusual data points using a configurable statistical trigger (PARAM-20). A trigger value shall be set for each data	One fixed Z-score for every data source assumes every signal behaves the same way.	T	Trigger required before operation

ID	Type	Requirement	Rationale	V	Note
		source and each anomaly type before that source is monitored in operation.			
UMR-043	Mandatory (revised)	When CRAFT flags an anomaly, it shall retrieve similar past anomalies from the knowledge base. The number returned and the similarity cutoff shall be configurable settings (PARAM-21).	Helps diagnose a new problem faster.	T	Same treatment as UMR-033
UMR-044	Mandatory (revised)	An anomaly at or above the high severity level defined in UMR-082 shall be sent to the assigned human role. CRAFT shall record whether it was acknowledged and shall keep the item open	“Serious” was undefined.	T	One severity scale

ID	Type	Requirement	Rationale	V	Note
		until a person closes it.			
UMR-045	Mandatory	Every detected anomaly and every human response to it shall be written to the audit record.	Keeps a full history for later review.	I	Unchanged
UMR-046	Mandatory (revised)	In phase 1, CRAFT shall produce reports in English, using the approved report templates. Reports include costing reports and reports that answer engineering questions or reasoning requests. French report generation is deferred to phase 2.	The template control applies. The report scope gives UMR-090 a defined object.	D	Report scope stated
UMR-047	Mandatory (revised)	Reports shall be built only from sources listed in the approved source register	Ensures reports use controlled information.	I	Bound to register and ceiling

ID	Type	Requirement	Rationale	V	Note
		(UMR-051). Every source in the register shall be at or below the Protected B ceiling in UMR-058.			
UMR-048	Mandatory	The written parts of a report shall be based on retrieved evidence and shall cite it, under UMR-006 and UMR-010.	Prevents invented report content.	T	Linked to grounding
UMR-049	Mandatory (revised)	CRAFT shall not allow a report to be marked final until an authorized reviewer records an approval.	A person must approve the final product.	T	Made enforceable
UMR-050	Mandatory	Final reports shall be held under version control and written to the audit record.	Supports document control and traceability.	I	Unchanged
UMR-051	Mandatory (revised)	CSA shall maintain an approved source register listing every	The review asked whether CRAFT must scrape documents	I, T	Five onboarding controls named in V3 from RC-01

ID	Type	Requirement	Rationale	V	Note
		repository, system and data feed CRAFT is permitted to read, with the classification and owner of each. CRAFT shall read only from sources in the register. In Phase 1 the register shall contain only the controlled Engineering Costing document corpus. Personal devices, individual OneDrive accounts and production CSA repositories shall not be added until each has passed a separately approved onboarding process that defines, at	from individual machines. The answer is no. Naming the five onboarding controls prevents a source being added later on the strength of convenience alone.		

ID	Type	Requirement	Rationale	V	Note
		minimum: (a) ownership; (b) access permissions; (c) data classification; (d) retention; (e) audit controls.			
UMR-052	Mandatory (revised)	CRAFT shall connect to SharePoint and Confluence through documented APIs, and to any other repository listed in the approved source register. These connectors shall be built in the source-onboarding phase, not in Phase 1. CSA's target state is that all repositories move to CSA SharePoint. CRAFT shall not depend on	Phase 1 uses a controlled corpus, so the connectors are not needed yet.	D	Deferred to the source-onboarding phase

ID	Type	Requirement	Rationale	V	Note
UMR-053	Mandatory (revised)	Confluence-specific behaviour that would prevent that move. CRAFT shall connect to the CSA financial system of record listed in the approved source register, for approved cost tasks. It is handled at Protected B under PBMM. Access shall be read-only under UMR-083.	Supports cost workflows without putting the system of record at risk.	T	In scope at Protected B
UMR-054	Mandatory (revised)	CRAFT shall connect to the mission design tools listed in the approved source register, which currently include STK and MATLAB.	Keeps engineering workflows connected.	D	Traced to UC-C8
UMR-055	Mandatory (revised)	CRAFT shall expose approved functions to	Controlled interfaces are essential in a system	I, T	Seven criteria separated

ID	Type	Requirement	Rationale	V	Note
		other systems through documented, version-controlled, secured interfaces. CRAFT may use REST, GraphQL, event-based messaging or another approved style. Every interface shall define: (a) its inputs; (b) its outputs; (c) its error handling; (d) its access rules; (e) its activity logging; (f) its compatibility rules; (g) its plan for retiring old versions.	built from many parts.		
UMR-056	Mandatory (revised)	CRAFT shall meet a fixed response-time target for each interactive request class, measured under the	A task the user starts and collects later is not measured by response time.	T	Report generation bound to UMR-090

ID	Type	Requirement	Rationale	V	Note
		approved load profile. The targets are: simple question, 5 seconds; retrieval-only question, 5 seconds; multi-step analysis, TBC. Report generation is not an interactive class; it runs on demand as a background task under UMR-090. Meeting a target shall not reduce answer quality, safety filtering or citation completeness.			
UMR-057	Mandatory (revised)	CRAFT shall measure and report availability during approved operating hours. The calculation shall be written down	End-to-end availability cannot be promised by the CRAFT application alone.	A, I	Owner and milestone

ID	Type	Requirement	Rationale	V	Note
		and approved, stating what counts as planned maintenance , what counts as an outside failure, and how incidents are classified. It shall separate what the CRAFT team controls from outside services, including Azure hosting, identity, model providers, data sources and network. The target shall be consistent with the Medium availability level in PBMM (PARAM-25).			
UMR-058	Mandatory (revised)	CRAFT shall handle information according to CSA-SEC-	A ceiling alone is not a control. Protected B covers the	T	OP-04 and OP-16 closed

ID	Type	Requirement	Rationale	V	Note
		POL-003. CRAFT shall be assessed and authorized at the Protected B / Medium integrity / Medium availability (PBMM) profile. The highest classification CRAFT is approved to hold is Protected B. CRAFT shall detect and reject the ingestion of any document or data feed carrying a classification above Protected B, and shall record every rejection in the audit record.	financial system of record and the official risk list.		
UMR-059	Mandatory (revised)	CRAFT shall support 500 active requests being processed at the same	The definition of concurrency is kept, because the original wording	T	Fixed by CSA decision

ID	Type	Requirement	Rationale	V	Note
		time, under the approved load profile. This is a fixed target.	could not be tested.		
UMR-060	Withdrawn	Merged into UMR-006. Identifier retained as a pointer. No separate requirement applies.	The old wording was a weaker duplicate of UMR-006.	—	Merged
UMR-061	Mandatory (revised)	CRAFT shall produce traceability reports in the approved export format, recording the evidence used, the retrieval path, the reasoning produced, the confidence given, any delegation, and the human approvals or rejections made, for every medium-risk and high-risk decision under UMR-082.	Supports audit and the ability to reconstruct a decision.	D	Retrieval path and delegation added in P13

ID	Type	Requirement	Rationale	V	Note
UMR-062	Mandatory (revised)	A user holding the Auditor role shall be able to export a time-stamped activity history in the approved export format, covering user requests, agent actions, retrieval paths, delegations, sources used, recommendations, decisions and later changes.	Supports governance reviews and investigations.	D	Scope widened in P13
UMR-063	Mandatory (new)	CSA shall maintain a Configurable Parameter Baseline Register (Annex B). Every configurable setting shall have an entry recording the parameter identifier, the interim	Without the register, “configurable” means “no control”.	I	Completes the three-layer model

ID	Type	Requirement	Rationale	V	Note
		baseline value, the owner, the evidence status and the next review date. A configurable setting shall not be left unset in an operational system.			
UMR-064	Mandatory (new)	CSA shall maintain an approved acceptance data set: questions, correct answers and source passages, held under version control. It shall be used to verify UMR-001, UMR-003, UMR-004, UMR-005, UMR-007, UMR-010, UMR-018 and UMR-094, and reviewed when the corpus or the model	Several requirements depend on this data set. Nothing required it to exist.	I	UMR-094 added to the list in P13

ID	Type	Requirement	Rationale	V	Note
UMR-065	Mandatory (revised)	changes. CRAFT shall authenticate every user through the approved CSA Azure identity service before any request is accepted. CRAFT shall enforce role-based access control using the role list in Section 3.4. Every user shall hold at least one role, and a role shall grant only the permissions listed for it. CRAFT shall record the role under which every action was performed.	A Protected B system cannot run on a single all-powerful account.	T	Role list, OP-13 closed
UMR-065.1	Mandatory (new)	CRAFT shall enforce separation of duties. The user who submits a request shall not be the	Self-approval makes an approval record worthless.	T	Added with the role list

ID	Type	Requirement	Rationale	V	Note
		user who approves it. The user who holds the Auditor role shall not hold the Administrator role.			
UMR-066	Mandatory (new)	CRAFT shall enforce, at query time, the access permissions held by the source repository for the requesting user, on every retrieval path, including agent-selected retrieval and file search under UMR-094. A user shall not receive, cite or see a passage from a document they are not permitted to read in the source system.	At a Protected B ceiling this is the primary control preventing disclosure. A retrieval tool that reads the corpus directly would otherwise bypass index-level trimming.	T	Scope widened to all paths in P13
UMR-067	Mandatory	CRAFT shall	A withdrawn	T	Chunk size

ID	Type	Requirement	Rationale	V	Note
	(new)	import, parse, split and index documents from sources in the approved source register. CRAFT shall read the classification label of every document before indexing it and apply the check in UMR-058. In the proof of concept, a document with no classification label shall be treated as Unclassified (PARAM-35), and every such document shall be recorded so the population can be reviewed. CRAFT shall re-index a document when the	document could otherwise be cited forever. Chunk size is separated from the request context floor.		setting, RC-02

ID	Type	Requirement	Rationale	V	Note
		source version changes, and shall remove it from the index and stop citing it when the source is deleted, withdrawn or superseded. Chunk size shall be a configurable setting (PARAM-39). Indexing delay shall be a configurable setting (PARAM-28).			
UMR-068	Mandatory (new)	CRAFT shall treat all retrieved document content as untrusted data, not as instructions. CRAFT shall detect and neutralise instruction-like content in retrieved passages. Defence shall be measured	An instruction hidden in a document can now steer both generated code and the choice of search tool, not only the wording of an answer.	T	Test set scope widened in P13

ID	Type	Requirement	Rationale	V	Note
		against the approved prompt-injection test set (PARAM-29), which shall include cases targeting code execution under UMR-093 and retrieval tool selection under UMR-094.			
UMR-069	Mandatory (new)	CRAFT shall encrypt data in transit and at rest using the approved Azure encryption services and algorithms, at the strength required by PBMM. Secrets and credentials shall be held in Azure Key Vault or an equivalent approved secret store, and shall not appear in prompts, logs, source	A Protected B ceiling raises the assurance level for credential handling.	T	Session workspace included

ID	Type	Requirement	Rationale	V	Note
UMR-070	Mandatory (new)	code, or the session workspace. CRAFT shall be hosted on Microsoft Azure, in a PBMM-assessed environment. CRAFT shall store and process CSA data only within Canadian Azure regions, including files uploaded by users and files created in a session workspace.	Hosting in Canadian Azure regions under a PBMM authorization satisfies the residency obligation.	I	Session workspace included
UMR-071	Withdrawn	Withdrawn by CSA decision. No Algorithmic Impact Assessment is required. Identifier retained. No requirement applies.	The Directive was moved to the background list.	—	See OP-14
UMR-072	Mandatory (new)	CRAFT shall identify personal information held in memory,	Protected B material routinely contains personal information.	T	Session workspace included

ID	Type	Requirement	Rationale	V	Note
		prompts, retrieved content, session workspaces and audit records, and shall provide a way to find and delete it on request, except where a record must be kept under UMR-073.			
UMR-073	Mandatory (new)	CRAFT records, including audit records, shall be retained and disposed of according to the applicable CSA records retention schedule. The schedule and the retention period are TBC.	Audit records cannot be kept forever without a schedule.	I	TBC, OP-07
UMR-074	Mandatory (new)	CRAFT shall be backed up and recoverable within the	Nothing covered data loss or recovery.	D	From Azure service tiers

ID	Type	Requirement	Rationale	V	Note
		approved recovery point and recovery time objectives (PARAM-30, PARAM-31). Recovery shall be demonstrated at the approved interval.			
UMR-075	Goal (phase 2)	In phase 2, CRAFT shall answer in the language of the question unless the user asks for another language. In phase 1, CRAFT answers in English.	Kept as a phase-2 goal so the need is not lost.	T	Phase-2 Goal
UMR-076	Withdrawn	Withdrawn by CSA decision. No French-language quality standard applies at this phase. Identifier retained.	CRAFT does not generate French output in phase 1.	—	OP-10 closed
UMR-077	Mandatory (new)	CRAFT shall support mission planning by	Mission planning support is named in the	T	Traced to UC-C8

ID	Type	Requirement	Rationale	V	Note
		retrieving planning evidence, summarising mission parameters and constraints, comparing options against recorded criteria, and citing the source of every parameter used. CRAFT shall not change mission parameters. A change to a mission parameter is high-risk under UMR-082.	scope.		
UMR-078	Mandatory (new)	CRAFT shall acquire telemetry and system health data from the feeds listed in the approved source register, at the configured interval	Anomaly detection needs data CRAFT is obliged to acquire.	D	Added

ID	Type	Requirement	Rationale	V	Note
UMR-079	Mandatory (new)	(PARAM-32). CRAFT shall deliver approval requests and anomaly reports through the approved notification channels, route each to the assigned role, and track delivery and acknowledgment.	Reports and approval requests need a delivery channel.	T	Added
UMR-080	Mandatory (new)	CRAFT shall monitor answer quality, grounding, confidence calibration and retrieval quality over time, and raise an alert when a measure falls below its baseline by more than the configured margin (PARAM-33). Monitoring shall cover each retrieval path	Quality degrades silently after a model or corpus change. With agent-selected retrieval, one path can degrade while the aggregate looks healthy.	T	Per-path monitoring added in P13

ID	Type	Requirement	Rationale	V	Note
UMR-081	Mandatory (new)	separately. CRAFT shall allow the language model to be replaced without changing application code. CRAFT shall re-run the acceptance set after any model change, including a change made by the model provider, and record the result.	Providers update models without CSA's involvement. The confidence method in UMR-007 no longer constrains this, because it does not need token probabilities.	T	Constraint removed by the UMR-007 decision
UMR-082	Mandatory (new)	CRAFT shall use one classification scheme for actions and findings, covering: (a) action reversibility — read-only, reversible, irreversible; (b) action risk tier — low, medium, high; (c) finding severity —	Every human-approval trigger depends on this.	I	Workspace export added in P13

ID	Type	Requirement	Rationale	V	Note
		low, medium, high, critical. The scheme shall state which combination s require human approval. High-risk actions shall include, at minimum: spending commitment s, changes to the official risk list, changes to mission parameters, code execution that writes outside the session workspace, export of session workspace content outside CRAFT, and any irreversible action. The scheme shall be held under change control.			

ID	Type	Requirement	Rationale	V	Note
UMR-083	Mandatory (new)	CRAFT shall have read-only access to the CSA financial system of record. CRAFT shall not write to it.	Read-only matters more now the source is confirmed in scope at Protected B.	T	Confirmed applicable
UMR-084	Mandatory (new)	CRAFT shall provide user guidance in English, covering what CRAFT can and cannot do, how to read a citation, how confidence scores should be interpreted, the limits of agent-generated analysis code, and how to read the retrieval path shown with an answer.	Users who do not understand the limits of the tool will over-trust it.	I	Retrieval path guidance added in P13
UMR-085	Mandatory (new)	The first operational release shall support 50 active requests being	Both figures are fixed requirements, one per phase.	T	From assumption OA-04

ID	Type	Requirement	Rationale	V	Note
		processed at the same time, under the approved load profile. This is a fixed target.			
UMR-086	Mandatory (new, split from UMR-010)	If CRAFT does not hold enough retrieved evidence to answer, it shall say so, ask a clarifying question, or decline to answer. CRAFT shall not answer from the model's own knowledge alone.	A distinct behaviour from measuring grounding.	T	Split from UMR-010
UMR-087	Mandatory (new, split from UMR-029)	For an irreversible action, CRAFT shall first obtain explicit approval from an authorized person. Before the action runs, CRAFT shall show what will change and what the effect will	Some actions cannot be undone.	T	Split from UMR-029

ID	Type	Requirement	Rationale	V	Note
UMR-088	Mandatory (new, split from UMR-029)	be. Any change CRAFT makes to a database, a system setting or program code shall use version control, a backup or an equivalent method, so the previously approved state can be restored.	This control was dropped from its mirror and is important enough to carry its own identifier.	T	Split from UMR-029
UMR-089	Mandatory (new, split from UMR-041)	Every anomaly report shall state the affected system, the time, the severity under UMR-082, the supporting evidence and the recommended next step.	Report content is a separate, testable rule from detection.	D	Split from UMR-041
UMR-090	Mandatory (revised)	A long-running task may run in the background. Report generation is a	Naming report generation and its scope makes the on-demand decision testable	D	Report scope stated

ID	Type	Requirement	Rationale	V	Note
		background task and runs on demand, covering costing reports and reports answering engineering questions or reasoning requests. For every background task, CRAFT shall show progress, allow the user to cancel it, and report when it is finished.	without a time figure.		
UMR-091	Mandatory (new)	The approved source register shall record, for every source, whether it holds real CSA data or simulated data. CRAFT shall label every retrieved passage and every citation that	Phase 1 runs on a controlled simulated corpus. A simulated cost estimate mistaken for a real one is a direct route to a bad decision.	T	Verification owner TBD, OP-21

ID	Type	Requirement	Rationale	V	Note
		comes from a simulated source, and shall not present simulated data as real mission evidence. Before any source holding real CSA data is added, the controls in UMR-051 and the classification controls in UMR-058, UMR-066 and UMR-067 shall be verified as working. The owner of that verification and the review at which it is held are TBD.			
UMR-092	Mandatory (revised)	CRAFT shall give each session an isolated workspace in which the agent may create files, such as charts,	The review asked whether each session gets its own filespace. The response sets the default to	T	Ephemeral default and export control added in V3 from RC-10

ID	Type	Requirement	Rationale	V	Note
		scripts, tables and draft documents. The workspace shall be ephemeral by default: it shall be destroyed when the session closes. Persistence beyond a session shall require an approved reason and shall follow the retention period in PARAM-41. A workspace shall be isolated from every other session, shall inherit the data classification of the material placed in it, and shall be held in Azure within Canadian regions. A user shall be able to list,	ephemeral and requires export control. An indefinitely retained workspace becomes an uncontrolled data store holding Protected B derivatives, and an uncontrolled export is the shortest path from a classified corpus to somewhere it should not be.		

ID	Type	Requirement	Rationale	V	Note
		download and delete the contents of their own workspace. Export of workspace content outside CRAFT shall be controlled, recorded in the audit record, and subject to the limits in PARAM-46. Files in a workspace shall not be treated as approved sources and shall not be indexed for retrieval unless added to the approved source register.			
UMR-093	Mandatory (new)	CRAFT may generate and run analysis code, such as a script for statistical inference over tabular data, to answer a	The review proposed that agents write and run scripts for data analysis. This is a genuine capability	T	See ASG-011

ID	Type	Requirement	Rationale	V	Note
		user request. All generated code shall run inside the containment described in ASG-011. CRAFT shall record the code, its inputs, its outputs and its exit status in the audit record, and shall present the code to the user alongside any result derived from it. A result produced by generated code shall carry the same citation and confidence obligations as any other answer. Code that would write outside the session workspace is a high-risk action under UMR-082.	gain and the highest-risk item in the commentary, so its containment is specified rather than assumed.		
UMR-094	Mandatory	The primary	The review	T	Added in

ID	Type	Requirement	Rationale	V	Note
	(new)	agent may select among authorized retrieval tools to answer a request, and may iterate. The authorized tool set shall be held in the tool registry under UMR-012 and shall include, at minimum: meaning-based search, keyword search, metadata filtering, and a parameterised file-search interface. The file-search interface shall accept a fixed set of named inputs such as path scope, filename pattern and	proposed that the agent choose how to search, including constructing commands such as grep or find. The capability is useful: a traceability question and a definition lookup want different strategies, and a single fixed pipeline serves one of them badly. Free-form shell strings are not acceptable, because the command space is unbounded, unauditable, and open to injection from ingested content. A parameterised interface preserves the capability and keeps		P13. Gap 1 closed, RC-17

ID	Type	Requirement	Rationale	V	Note
		content pattern. CRAFT shall not permit an agent to compose or execute free-form shell or command-line strings. The normalised, fused default path in UMR-004 shall remain available and shall be the default where the agent expresses no preference. The number of retrieval iterations per request shall be a configurable setting (PARAM-43). Every tool selection shall be recorded in the retrieval path under ARR-010, and every path shall be subject to UMR-066.	the boundary.		

ID	Type	Requirement	Rationale	V	Note
UMR-095	Mandatory (new)	CRAFT shall provide a conversational interface. Each session shall have one primary agent that persists for the life of the session, holds the episodic memory defined in UMR-014, and is bound to the session workspace defined in UMR-092. The session, its agent context, and its workspace shall end together. Session lifetime and idle timeout shall be configurable settings (PARAM-44). A session shall be tied to one authenticated user under UMR-065,	The review's stated direction was a chat interface with a persistent agent rather than one-shot model calls augmented with retrieval. V2 covered free-text questions and workspaces but never stated the interaction model, so the persistence, ownership and lifetime of a session were undefined. Ending the session, the agent context and the workspace together prevents orphaned state holding Protected B derivatives.	T	Added in P13. Gap 2 closed, RC-18

ID	Type	Requirement	Rationale	V	Note
		and session state shall not be readable by any other session.			

4.2 Agent requirements (detail)

Every tag mirrors one or more requirements in Section 4.1. Where the two appear to differ, Section 4.1 governs.

4.2.1 How agents are built

Tag	Type	Requirement	Mirrors	V
AAR-001	Mandatory (revised)	Each session shall have one accountable primary agent. Fixed agent roles shall not be imposed, and multiple agents shall not be used by default.	UMR-011, UMR-095	I
AAR-002	Mandatory (revised)	An agent shall use only the tools listed in the approved tool registry for its task, including retrieval tools.	UMR-012	T
AAR-003	Mandatory (revised)	Agents shall run within a configurable budget covering time, tokens, tool calls, reasoning steps, retrieval iterations,	UMR-013, UMR-030	T

Tag	Type	Requirement	Mirrors	V
		delegated subtasks, estimated cost, and code execution resources. When the budget is used up, the agent shall stop safely, return any useful partial result, record the reason, and ask the user what to do next.		
AAR-004	Mandatory (revised)	Agents shall have episodic, semantic and procedural memory as defined in UMR-014, each with a defined retention period and deletion path. Semantic memory entries shall be source-backed. Procedural memory shall be approved and version-controlled.	UMR-014	I, T
AAR-005	Mandatory	Agent-to-agent messages shall include the agent identifier, the action, the input, the	UMR-015	I

Tag	Type	Requirement	Mirrors	V
		output, a confidence score and a timestamp.		
AAR-006	Mandatory (revised)	An agent shall be able to be started or stopped while the platform continues to serve requests. Requests running elsewhere shall not fail.	UMR-016	D
AAR-007	Mandatory (revised)	An agent shall not read, change or delete data, system functions or system settings outside its registered tool permissions.	UMR-017	T
AAR-008	Mandatory (new)	Agent memory shall be isolated per user and per data classification. One session's memory shall not be readable in another session.	UMR-014, UMR-058, UMR-095	T
AAR-009	Mandatory (new)	An agent shall not modify its own instructions, prompts, tool permissions, code, or	UMR-028	T

Tag	Type	Requirement	Mirrors	V
AAR-010	Mandatory (new)	procedural memory. Changes go through UMR-028. Where the primary agent delegates a subtask, it shall record the reason and the bounds, shall validate the returned result before use, and shall integrate it into the final response. The primary agent remains accountable for the response given to the user.	UMR-011	T

4.2.2 How agents plan, search and answer

Tag	Type	Requirement	Mirrors	V
ARR-001	Mandatory	CRAFT shall break a complex request into steps and record the plan.	UMR-018	T
ARR-002	Mandatory (revised)	CRAFT shall retrieve using keyword and vector search behind a common interface, normalising, fusing and	UMR-004, UMR-005	I, T

Tag	Type	Requirement	Mirrors	V
		deduplicating the results. Appending the raw top results of each method is prohibited. Weights shall be configurable and version-controlled. Phase 1 baseline: 0.7 semantic, 0.3 lexical.		
ARR-003	Mandatory (revised)	Every answer shall name, for each fact used, the source document, its version and the page or section. Where the format has no such markers, the passage locator shall be given.	UMR-006	D
ARR-004	Mandatory (revised)	Every answer shall carry a confidence score between 0 and 1, computed from evidence coverage, citations, source traceability, source agreement, retrieval quality and	UMR-007	D

Tag	Type	Requirement	Mirrors	V
		unsupported-claim detection. Token-probability scoring shall not be used.		
ARR-005	Mandatory (revised)	An output with a confidence score below the approved threshold for its risk tier shall be routed to the Reviewer role. Interim baseline value: 0.6.	UMR-008, UMR-065	T
ARR-006	Mandatory (revised)	If the first search does not meet the sufficiency condition, the agent shall search again with adjusted terms or another authorized retrieval tool, up to the configured iteration limit.	UMR-009, UMR-094	T
ARR-007	Mandatory (revised)	All generated content shall be based on retrieved evidence. The pass score, test method and checking tool shall be written down before acceptance	UMR-010	T

Tag	Type	Requirement	Mirrors	V
ARR-008	Mandatory (new)	testing. If the evidence is not sufficient, the agent shall say so, ask a clarifying question, or decline to answer. It shall not answer from the model's own knowledge alone.	UMR-086	T
ARR-009	Mandatory (new)	Where an agent derives a result by running generated code, the answer shall state that code was used, show the code, and cite the data the code read.	UMR-093, UMR-006	D
ARR-010	Mandatory (new)	CRAFT shall record the retrieval path for every answer: which retrieval tools were used, in what order, with what parameters, and how many iterations were run. The retrieval path shall be available to the user on request, shall accompany	UMR-094, UMR-027, UMR-061	T

Tag	Type	Requirement	Mirrors	V
		every approval request, and shall be written to the audit record.		

4.2.3 Human review and approval

Tag	Type	Requirement	Mirrors	V
HITL-001	Mandatory (revised)	CRAFT shall obtain approval from an authorized person before any action classified as high-risk under UMR-082.	UMR-021, UMR-082	T
HITL-002	Mandatory (revised)	High-risk actions are those listed in the scheme under UMR-082, including spending commitments, changes to the official risk list, changes to mission parameters, code execution writing outside the session workspace, workspace export outside CRAFT, and any irreversible action.	UMR-082	I
HITL-003	Mandatory (revised)	Every approval request shall show the	UMR-022, ARR-010	D

Tag	Type	Requirement	Mirrors	V
		reasoning, the confidence score, the source documents used and the retrieval path.		
HITL-004	Mandatory (revised)	A person holding the Approver role shall be able to approve, reject or edit a recommendation before it becomes final.	UMR-023, UMR-065	D
HITL-005	Mandatory (revised)	If a recommendation is rejected, the agent may try again using the reviewer's correction, up to the configured retry limit.	UMR-024	T
HITL-006	Mandatory (revised)	If no decision is recorded within the configured timeout, the request shall be escalated to the Backup approver, then cancelled and flagged if still undecided. If no Backup approver is named, the request shall stay open and	UMR-025	T

Tag	Type	Requirement	Mirrors	V
		be flagged, not cancelled. Interim baseline timeout: 24 hours.		
HITL-007	Mandatory	Every approval, rejection or edit shall be logged with who did it, what was decided, when and why.	UMR-026	I
HITL-008	Mandatory (new)	A request reporting a high-severity operational anomaly shall not be cancelled by timeout. It shall remain open until a person closes it.	UMR-025, UMR-044	T

4.2.4 Safety and oversight

Tag	Type	Requirement	Mirrors	V
ASG-001	Mandatory (revised)	CRAFT shall filter generated content against the approved content policy list, measured against the approved adversarial test set.	UMR-020	T
ASG-002	Mandatory (revised)	Every agent action shall be classified as read-only, reversible or irreversible	UMR-029, UMR-087, UMR-088	T

Tag	Type	Requirement	Mirrors	V
		under UMR-082. A reversible action shall be checked before it runs, shall save the earlier state or a version marker, shall be logged, and shall be undoable. An irreversible action shall require explicit human approval first, with a preview of what will change.		
ASG-003	Mandatory (revised)	CRAFT shall write every agent decision and action to append-only storage, including retrieval paths, delegations, workspace exports and code execution. No platform role, including administrator, shall be able to change or delete a record.	UMR-027	T
ASG-004	Mandatory (revised)	CRAFT shall limit resource use per user, per session, per task and per agent, covering	UMR-030	T

Tag	Type	Requirement	Mirrors	V
		tokens, concurrent tasks, estimated cost, tool calls, call rate, retrieval iterations, delegated subtasks, and code execution time and memory. Soft limits shall warn and slow new work. Hard limits shall stop the task safely.		
ASG-005	Mandatory (revised)	If an agent repeats an action matching the approved repetition test with no new information, CRAFT shall stop the task and escalate. Interim baseline value: 3.	UMR-019	T
ASG-006	Mandatory (revised)	Every version of an agent's model settings, prompts, tool permissions and procedural memory shall be version-controlled, reviewed and approved before operational use.	UMR-028	I

Tag	Type	Requirement	Mirrors	V
ASG-007	Mandatory (new)	Retrieved content shall be treated as data, never as instructions. Instruction-like content shall be detected and neutralised, measured against the approved prompt-injection test set, including cases targeting code execution and retrieval tool selection.	UMR-068	T
ASG-008	Mandatory (new)	The audit store shall be held in an Azure service configured with immutable, time-based retention, so that no role, including the administrator role, can change or delete a record.	UMR-027, UMR-065	T
ASG-009	Mandatory (new)	CRAFT shall refuse to index any document whose classification is above the Protected B ceiling, shall not retain the content of a	UMR-058, UMR-067	T

Tag	Type	Requirement	Mirrors	V
		refused document, and shall write the refusal to the audit record.		
ASG-010	Mandatory (new)	CRAFT shall apply the controls required by the PBMM profile for Protected B information, and shall record the mapping between each PBMM control and the requirement that implements it.	UMR-058, UMR-069, UMR-070	I
ASG-011	Mandatory (new)	Generated code shall run in a contained environment that: (a) has no network access by default; (b) can read only the data explicitly passed to it; (c) can write only inside the session workspace; (d) runs under the limits in PARAM-42; (e) has no access to secrets, credentials or the audit store;	UMR-093, UMR-069, UMR-082	T

Tag	Type	Requirement	Mirrors	V
		(f) is destroyed after the task. Every execution shall be recorded with its code, inputs, outputs and exit status. An attempt to act outside the containment shall be stopped and flagged to the Security authority role.		
ASG-012	Mandatory (new)	Agent retrieval shall be constrained at the tool boundary. CRAFT shall expose only the retrieval tools listed in the approved tool registry. CRAFT shall not expose a shell, a command-line interpreter, or any interface accepting a free-form command string. The file-search tool shall accept only named parameters and shall be confined to the	UMR-094, UMR-012, UMR-066	T

Tag	Type	Requirement	Mirrors	V
		path scope of sources in the approved source register. Permission trimming under UMR-066 shall apply to every retrieval tool, including direct file search. Every tool selection, its parameters, and every blocked attempt shall be written to the audit record and made available to the Security authority role.		

5. Architecture

5.1 Status

No Architecture Design Document exists. CSA-CRAFT-AD-0001 Draft A1 contains a cover page and two framing paragraphs, with no interfaces, data flow, trust boundaries, deployment view or decision records. RD-01 is treated as not available. Section 5.2 is an allocation view only, not an architecture, and nothing in it is approved for procurement.

5.2 Building blocks implied by the requirements

Block	Purpose	Requirements it must satisfy
Ingestion and parsing	Imports documents, extracts text and metadata, chunks, checks classification	UMR-051, UMR-058, UMR-067, ASG-009
Knowledge store	Search index for meaning-based search, and a	UMR-004, UMR-067

Block	Purpose	Requirements it must satisfy
	knowledge graph	
Retrieval engine	Keyword and vector retrieval behind one interface, normalisation, fusion, deduplication, permission trimming	UMR-004, UMR-005, UMR-066
Retrieval tool layer	Authorized retrieval tools exposed to the agent, including parameterised file search	UMR-094, UMR-012, ASG-012
Primary agent and delegation	Plans tasks, calls tools, writes answers, delegates bounded subtasks, validates results	UMR-011 to UMR-019, UMR-095, all AAR
Memory services	Episodic, semantic and procedural memory, with curation, approval, retention and isolation	UMR-014, UMR-072, UMR-073, AAR-004, AAR-008
Session workspace	Ephemeral per-session filesystem for agent-created files, with controlled export	UMR-092, UMR-069, UMR-070, UMR-072
Code execution sandbox	Contained environment for generated analysis code	UMR-093, ASG-011, UMR-030
Grounding and confidence service	Binds answers to evidence, computes confidence from six named inputs, detects unsupported claims	UMR-006 to UMR-010, UMR-086
Human review workflow	Shows evidence and retrieval path, records decisions, handles timeout and escalation	UMR-021 to UMR-026, all HITL
Audit and logging	Append-only record covering actions, retrieval paths, delegations, exports and code execution	UMR-015, UMR-027, UMR-061, UMR-062, ASG-008
Safety services	Content filtering, action classification, injection defence, resource governor, tool and code containment	UMR-020, UMR-029, UMR-030, UMR-068, UMR-082, all ASG
Integration layer	Controlled connections to	UMR-052 to UMR-055, UMR-

Block	Purpose	Requirements it must satisfy
	the financial system, STK, MATLAB, data feeds, and later repositories	078, UMR-083
Identity and access	Azure identity, user role model, Protected B ceiling enforcement, session ownership	UMR-058, UMR-065, UMR-065.1, UMR-066, UMR-095
Interface layer	Conversational session interface, progress, cancellation, guidance	UMR-001, UMR-055, UMR-079, UMR-084, UMR-090, UMR-095
Monitoring	Availability, performance, per-path retrieval quality, drift, recovery	UMR-056, UMR-057, UMR-074, UMR-080

Deployment infrastructure

Azure is the deployment infrastructure for CRAFT. The on-premises infrastructure estimate in Appendix A is withdrawn in full. Those figures do not convert: an on-premises estimate counts machines, uplinks and disks; an Azure deployment is specified by service tiers, instance families, regional capacity and consumption. Azure sizing shall be derived from first principles in RD-01.

Workload assumptions retained from Appendix A:

Assumption	Value	Where it is carried
Phase-1 concurrency	50 active requests at the same time	UMR-085
Target concurrency	500 active requests at the same time	UMR-059
Availability	99.5 percent during CSA operating hours	PARAM-25, UMR-057

Azure service classes implied by the requirements. An allocation list, not a design. No tier, size, quantity or cost is stated, because none has been derived.

Building block	Azure service class needed	Driven by
Ingestion and parsing	Compute for background workers; document parsing service	UMR-067
Knowledge store	Managed vector index; managed graph or relational store	UMR-004, UMR-067

Building block	Azure service class needed	Driven by
Retrieval engine and tool layer	Search service supporting keyword and vector retrieval, with permission trimming and a constrained file-search facility	UMR-004, UMR-005, UMR-066, UMR-094
Model inference	Managed model endpoint in a Canadian region, with a substitution path	UMR-070, UMR-081
Primary agent runtime	Application compute with autoscale and session affinity	UMR-095, UMR-059, UMR-085
Memory services	Cache and persistent store, isolated per user and per classification	UMR-014, AAR-008
Session workspace	Isolated ephemeral file storage with lifecycle policy and export control	UMR-092
Code execution sandbox	Isolated container runtime with no default network egress	UMR-093, ASG-011
Human review workflow	Workflow or queue service with state persistence	UMR-021 to UMR-026
Audit and logging	Immutable storage with time-based retention	UMR-027, UMR-073, ASG-008
Identity and access	Azure identity, role-based access control	UMR-065, UMR-065.1
Secrets and encryption	Key Vault or approved equivalent; platform encryption at PBMM strength	UMR-069
Backup and recovery	Geo-redundant backup within Canadian regions	UMR-074, UMR-070
Monitoring	Platform monitoring, log analytics, per-path quality alerting	UMR-057, UMR-080

All of these shall sit inside a PBMM-assessed Azure landing zone in Canadian regions only, as required by UMR-070. The definitive topology, service tiers, redundancy model, scaling policy and cost position shall be fixed in RD-01 under ADR-08 and ADR-18.

5.3 Required content of the Architecture Design Document

Every architecture decision shall be recorded as an Architecture Decision Record (ADR), with a Satisfies field naming the requirement identifiers it implements and a Sets baseline for field naming the Annex B parameters it fixes. A DESIGN DECISION without both fields is not complete.

§	Section	Minimum content	Traces to
1	Introduction and definitions	Purpose, readership, acronym table, relationship to CSA-CRAFT-RD-0001	—
2	Applicable documents	AD-01 to AD-08, plus hosting and security standards	AD-01 to AD-08
3	Architecture drivers	The constraints that shape the design	UMR-058, UMR-065 to UMR-073
4	Context view	External systems, users, data in and out, trust boundaries	UMR-051 to UMR-055, UMR-078
5	Functional view	Allocation of every requirement identifier to a component	all
6	Information view	Document model, chunking, metadata, classification labels, graph schema	UMR-058, UMR-067, UMR-073
7	Retrieval design	Keyword and vector retrieval, common interface, normalisation, fusion, deduplication, threshold validation, reranking, permission trimming, evaluation harness	UMR-004, UMR-005, UMR-064, UMR-066
8	Agent design and delegation	Primary agent model, tool registry, delegation criteria	UMR-011 to UMR-019, UMR-095, AAR-001 to AAR-010

§	Section	Minimum content	Traces to
		and bounds, validation and integration, planner, budget enforcement, loop breaker	
9	Retrieval tool design	Authorized tool set, parameterised file- search interface, iteration control, path recording	UMR-094, ARR-010, ASG-012
10	Memory design	Episodic, semantic and procedural implementation, curation and approval workflow, isolation, retention, deletion	UMR-014, UMR-072, UMR-073, AAR-004
11	Grounding and confidence design	Evidence binding, citation generation, six-input confidence model, source agreement, unsupported-claim detection, calibration	UMR-006 to UMR- 010, UMR-086, ARR- 003 to ARR-008
12	Human review workflow engine	State model, queues, roles, timeout, backup approver, preview rendering	UMR-021 to UMR- 026, HITL-001 to HITL-008
13	Audit and logging mechanism	Record schema, append-only store, integrity proof, export, retention, retrieval path and delegation records	UMR-015, UMR-027, UMR-061, UMR-062, ASG-003, ASG-008
14	Safety architecture	Action classification, reversibility, content filter, injection defence, resource	UMR-020, UMR-029, UMR-030, UMR-068, UMR-082, ASG-001 to ASG-012

§	Section	Minimum content	Traces to
		governor, tool and code containment	
15	Integration architecture	Connector pattern, authentication, rate limits, read-only enforcement, source-onboarding controls	UMR-051 to UMR-055, UMR-078, UMR-083
16	Deployment topology	Azure environments in a PBMM-assessed landing zone, Canadian regions, service tiers, sizing, scaling, backup, cost position	UMR-057, UMR-059, UMR-070, UMR-074, UMR-085
17	Performance and capacity design	Interactive request classes, background execution, session affinity, queueing, load model	UMR-056, UMR-059, UMR-090, UMR-095
18	Security architecture	Azure identity, user role model, separation of duties, encryption, Key Vault, Protected B enforcement, sandbox and tool threat model, PBMM control mapping	UMR-017, UMR-058, UMR-065, UMR-066, UMR-069, UMR-093, UMR-094, ASG-010 to ASG-012
19	Operability	Monitoring, per-path drift detection, model change management, incident handling	UMR-080, UMR-081
20	Architecture Decision Records	One per decision, with Satisfies, Sets baseline for, Alternatives, Evidence	all
21	Requirement-to-component	Every UMR, AAR, ARR, HITL and ASG	all

§	Section	Minimum content	Traces to
	traceability matrix	mapped to a component and an ADR	
A	Annex A	Baseline parameter values with supporting evidence	Annex B of this document
B	Annex B	Open design decisions still to be tested	Annex E of this document

Minimum set of Architecture Decision Records:

ADR	Decision	Satisfies
ADR-01	Retrieval methods, common interface, normalisation, fusion and deduplication	UMR-004, UMR-005, ARR-002
ADR-02	Permission trimming across every retrieval path	UMR-066, UMR-058, ASG-012
ADR-03	Primary agent model, delegation criteria and accountability. Records the decision that closed OP-30	UMR-011, UMR-012, AAR-001, AAR-010
ADR-04	Budget enforcement point and stop behaviour	UMR-013, UMR-030, AAR-003, ASG-004
ADR-05	Memory tier implementation, curation workflow and retention	UMR-014, UMR-073, AAR-004
ADR-06	Human review workflow engine and approval state model	UMR-021 to UMR-026, HITL-001 to HITL-008
ADR-07	Audit logging mechanism and integrity proof	UMR-027, UMR-062, ASG-003
ADR-08	Azure PBMM landing zone, Canadian region selection and residency	UMR-057, UMR-070, UMR-074, UMR-085
ADR-09	Language model sourcing and substitution path	UMR-028, UMR-081
ADR-10	Confidence computation from the six named inputs, and its calibration	UMR-007, UMR-010, ARR-004

ADR	Decision	Satisfies
ADR-11	Prompt-injection defence, including code and tool-selection cases	UMR-068, ASG-007
ADR-12	Anomaly detection method and trigger tuning	UMR-041, UMR-042, UMR-078
ADR-13	Azure immutable audit store, binding on the Administrator role	UMR-027, UMR-065, ASG-008
ADR-14	Classification detection at ingestion, and handling of unlabelled documents	UMR-058, UMR-067, ASG-009
ADR-15	User role model, separation of duties, and Azure identity integration	UMR-065, UMR-065.1
ADR-16	PBMM control implementation and the control-to-requirement mapping	UMR-058, UMR-069, UMR-070, ASG-010
ADR-17	Background task execution, progress reporting and cancellation	UMR-056, UMR-090
ADR-18	Azure sizing derived from the workload assumptions, service tiers and cost position	UMR-059, UMR-070, UMR-074, UMR-085
ADR-19	Session workspace implementation, ephemeral default, isolation, export control	UMR-092, UMR-069, UMR-072
ADR-20	Code execution sandbox: runtime, containment boundary, resource limits and audit	UMR-093, ASG-011, UMR-082
ADR-21	Chunking strategy and its relationship to the request context floor	UMR-003, UMR-067, PARAM-39
ADR-22	Retrieval tool set, parameterised file-search interface, and iteration control	UMR-094, ARR-010, ASG-012

ADR	Decision	Satisfies
ADR-23	Session lifecycle: creation, persistence, idle timeout, and joint termination of agent context and workspace	UMR-095, UMR-092, PARAM-44

Annex A — Identifier mapping

Old identifier	New identifier	Reason
UMR-010	UMR-010 + UMR-086	Insufficient-evidence behaviour separated
UMR-029	UMR-029 + UMR-087 + UMR-088	Irreversible-action approval and change control separated
UMR-041	UMR-041 + UMR-089	Anomaly report content separated
UMR-056	UMR-056 + UMR-090	Background task behaviour separated
UMR-051, UMR-052	UMR-051 (reissued), UMR-052	Source register separated from connectors
UMR-060	merged into UMR-006	Weaker duplicate. Identifier retained as a pointer
UMR-065	UMR-065 + UMR-065.1	Separation of duties separated
UMR-071, UMR-076	withdrawn	Identifiers retained as pointers
UC-C1 (<i>Dublin Core</i>)	UC-C7	Identifier collision resolved
—	UC-C8, UC-C9	Written to give orphaned requirements a stated demand
UMR-014 tier names	episodic, semantic, procedural	Renamed in V2 , definitions tightened in P13. Identifier unchanged
—	UMR-092, UMR-093, ASG-011	V2 : session workspace, code execution, containment
—	UMR-094 (new)	Agent-selected retrieval, V3 Gap 1

Old identifier	New identifier	Reason
—	UMR-095 (new)	Session-scoped primary agent, V3 Gap 2
—	AAR-010 (new)	Delegation accountability
—	ARR-010 (new)	Retrieval path recording and disclosure
—	ASG-012 (new)	Retrieval tool containment

All other identifiers are unchanged. No identifier was deleted.

Annex B — Configurable Parameter Baseline Register

Required by UMR-063.

Parameter	Setting	Requirement	Evidence status
PARAM-01	Language detection accuracy	UMR-001	Phase 1 answers in English only
PARAM-02	Minimum total request context	UMR-003	A floor for model compatibility, not a target. Larger windows permitted; retrieval stays selective
PARAM-03	Result count, filter threshold, fusion method, primary quality measure	UMR-004	0.72 may be adopted only once measured against the acceptance set
PARAM-04	Retrieval method weights	UMR-005	Set by review decision. Configurable in later phases behind the common interface
PARAM-05	Retry limit after insufficient retrieval	UMR-009	Engineering judgement
PARAM-06	Confidence gate per risk tier	UMR-008, ARR-005	Untested
PARAM-07	Grounding pass score per risk tier	UMR-010	Untested
PARAM-08	Content filter block rate and false-block rate	UMR-020	Requires an adversarial test set

Parameter	Setting	Requirement	Evidence status
PARAM-09 to PARAM-13	Budget limits: time, tokens, tool calls, reasoning steps, cost	UMR-013, UMR- 030	Carried from the original draft
PARAM-14	Repetition limit	UMR-019, ASG- 005	Untested
PARAM-15	Retry limit after rejection	UMR-024, HITL- 005	Engineering judgement
PARAM-16	Approval timeout per risk tier	UMR-025, HITL- 006	Untested
PARAM-17	Cost estimate accuracy	UMR-031	Requires a back-test
PARAM-18	Mission comparison method and weights	UMR-032	Appendix B, UC-C4
PARAM-19	Comparison mission count and cutoff	UMR-033	Untested
PARAM-20	Anomaly statistical trigger, per source	UMR-042	Untested
PARAM-21	Similar past anomaly count and cutoff	UMR-043	Engineering judgement
PARAM-22 to PARAM-24	Withdrawn. Response-time targets are fixed in UMR-056	UMR-056	Withdrawn, OP-12
PARAM-25	Availability target	UMR-057	Assumption OA-03. Consistent with PBMM Medium availability
PARAM-26	Withdrawn. Concurrency is fixed in UMR- 059	UMR-059	Withdrawn, OP-12
PARAM-27	Memory	UMR-014	Waiting on the records schedule,

Parameter	Setting	Requirement	Evidence status
	retention per tier		OP-07
PARAM-28	Indexing delay	UMR-067	Requires sizing analysis
PARAM-29	Prompt-injection defence rate	UMR-068	Test set must include code and tool-selection cases
PARAM-30, PARAM-31	Recovery point and recovery time objectives	UMR-074	From Azure service tiers and PBMM Medium availability
PARAM-32	Telemetry acquisition interval	UMR-078	Per data feed
PARAM-33	Quality drift alert margin	UMR-080	Requires a quality baseline, per retrieval path
PARAM-34	Withdrawn. Phase-1 concurrency is fixed in UMR-085	UMR-085	Withdrawn, OP-12
PARAM-35	Default handling of an unlabelled document	UMR-058, UMR-067	Proof of concept only. Review before production authorization, OP-17
PARAM-36	Role assignment review interval	UMR-065	Engineering judgement
PARAM-37	Named Backup approver for each Approver	UMR-025, HITL-006	Not assigned at the proof-of-concept stage, OP-19
PARAM-38	Source data type: real or simulated, per source	UMR-091	Set by CSA decision
PARAM-39	Document chunk size and overlap	UMR-067, UMR-003	Separated from the request context floor. Must be set by retrieval testing
PARAM-40	Procedural memory implementation	UMR-014	Agents may not write to these

Parameter	Setting	Requirement	Evidence status
PARAM-41	Session workspace retention where persistence is approved	UMR-092	Default set in P13. A persistence value is needed only if persistence is approved
PARAM-42	Code execution time and memory limits	UMR-093, ASG-011	Must be set before code execution is enabled, OP-31
PARAM-43	Retrieval iterations per request	UMR-094, ARR-006	New in P13. Without a limit, agent-chosen retrieval can loop at cost
PARAM-44	Session lifetime and idle timeout	UMR-095	New in P13. Governs when the agent context and workspace are destroyed
PARAM-45	Delegation criteria and thresholds	UMR-011, AAR-010	New in P13. Makes “demonstrably improves quality, speed, isolation or capability” measurable rather than asserted
PARAM-46	Session workspace export limits	UMR-092, UMR-082	New in P13. Size, format and approval threshold for taking content out of CRAFT

Annex C — Traceability

Method. Requirements serving a user demand trace to a use case. Requirements that are cross-cutting controls trace to an architecture component, a DESIGN DECISION and a verification procedure, in the register at C.5.

C.1 Use cases

Use case	Title	Actor
UC-A1	Systems engineering documentation ingestion and concept deduction	Systems engineer
UC-B1	Global space technology benchmarking and horizon scanning	Systems engineer
UC-C1	Natural language technical query	CSA engineer
UC-C2	Verifiable question answering with deep	Finance engineer, systems engineer

Use case	Title	Actor
	citations, and cost estimation	
UC-C3	Tabular data transformation to SQL, and risk identification and flagging	Systems engineer, risk manager
UC-C4	Parametric mission distance calculation, and anomaly detection	Operations engineer, mission control
UC-C5	Automated report compilation	Systems engineer, project manager
UC-C6	Enterprise cognitive synthesis layer for the engineering lifecycle	Systems engineer, cost analyst, scientist, data governance officer
UC-C7	Dublin Core metadata extraction	Systems engineer
UC-C8	Mission planning support	Systems engineer, mission analyst
UC-C9	Governance and audit reporting	Auditor, security authority, technical authority

C.2 Requirement to use case coverage

Requirement group	Traced to
UMR-001, UMR-002, UMR-009	UC-C1
UMR-004	UC-C1, UC-C2, UC-C3, UC-C4, UC-C6, UC-C7
UMR-005 to UMR-008, UMR-010, UMR-011, UMR-018	UC-C1 to UC-C4, UC-C6
UMR-013 to UMR-017, UMR-019, UMR-020	UC-C1 to UC-C4
UMR-021 to UMR-030	UC-C2 to UC-C5, UC-C8
UMR-031 to UMR-035	UC-C2
UMR-036 to UMR-040	UC-C3
UMR-041 to UMR-045	UC-C4
UMR-046 to UMR-050	UC-C5
UMR-052, UMR-053	UC-C5
UMR-054, UMR-077	UC-C8
UMR-055 to UMR-059	UC-C1 to UC-C5
UMR-061, UMR-062, UMR-026, UMR-027,	UC-C9

Requirement group	Traced to
UMR-045, UMR-050	
UMR-062, UMR-065, UMR-066, UMR-075	UC-C6
UMR-002, UMR-006, UMR-067	UC-C7
UMR-051, UMR-067, UMR-068	UC-A1
UMR-051, UMR-068, UMR-086, UMR-091	UC-B1
UMR-092, UMR-093	UC-C3, UC-C4
UMR-094	UC-A1, UC-B1, UC-C1, UC-C3, UC-C7 — documentation ingestion, horizon scanning, free-text query and metadata extraction each need a different search strategy, which is the demand behind agent-selected retrieval
UMR-095	UC-C1, UC-C6 — the natural language query and cognitive synthesis use cases both describe multi-turn interaction, which requires a persistent session

C.5 Cross-cutting requirements register

Requirement	Component (Section 5.2)	ADR	V
UMR-003	Grounding and confidence service; ingestion	ADR-10, ADR-21	T
UMR-012	Retrieval tool layer; primary agent; safety services	ADR-03, ADR-22	T
UMR-063	Configuration control, Annex B	ADR-04	I
UMR-064	Retrieval and grounding evaluation harness	ADR-01, ADR-10	I
UMR-065, UMR-065.1	Identity and access	ADR-15	T
UMR-066	Retrieval engine; retrieval tool layer	ADR-02	T
UMR-067	Ingestion and parsing	ADR-14, ADR-21	T
UMR-068	Safety services	ADR-11	T

Requirement	Component (Section 5.2)	ADR	V
UMR-069	Secrets and encryption	ADR-08, ADR-16	T
UMR-070	Deployment topology	ADR-08	I
UMR-072	Memory services; session workspace; audit	ADR-05, ADR-19	T
UMR-073	Audit and logging	ADR-07	I
UMR-074	Backup and recovery	ADR-08, ADR-18	D
UMR-078	Integration layer	ADR-12	D
UMR-079	Interface layer; human review workflow	ADR-06	T
UMR-080	Monitoring	ADR-10	T
UMR-081	Model inference	ADR-09	T
UMR-082	Safety services	ADR-06, ADR-19, ADR-20	I
UMR-083	Integration layer	ADR-14	T
UMR-084	Interface layer	—	I
UMR-085, UMR-059	Deployment topology; agent runtime	ADR-18	T
UMR-088	Safety services	ADR-04	T
UMR-091	Ingestion; approved source register	ADR-14	T
UMR-092	Session workspace	ADR-19	T
UMR-093	Code execution sandbox	ADR-20	T
UMR-094	Retrieval tool layer	ADR-22	T
UMR-095	Primary agent runtime; interface layer	ADR-23	T
AAR-008	Memory services	ADR-05	T
AAR-009	Primary agent; memory services	ADR-03, ADR-05	T
AAR-010	Primary agent and	ADR-03	T

Requirement	Component (Section 5.2)	ADR	V
	delegation		
ARR-009	Interface layer;	ADR-20	D
	grounding service		
ARR-010	Retrieval tool layer;	ADR-22, ADR-07	T
	audit and logging		
ASG-008	Audit and logging	ADR-13	T
ASG-009	Ingestion and	ADR-14	T
	parsing		
ASG-010	Security architecture	ADR-16	I
ASG-011	Code execution	ADR-20	T
	sandbox		
ASG-012	Retrieval tool layer;	ADR-22, ADR-02	T
	safety services		

C.6 Completeness check

Every requirement identifier in Section 4 traces to either a use case or the cross-cutting register. No mandatory requirement is orphaned. UMR-060, UMR-071 and UMR-076 are withdrawn and require no trace.

Completeness is not correctness. Whether each mapping is right is confirmed at the Requirements Review by the people who own each use case. UC-C8 and UC-C9 were written by this review rather than by the engineers who will use them (OP-29).

Annex D — Verification

Every requirement carries a verification method in the V column. The verification plan shall record, for each requirement: the method, the test procedure identifier, the acceptance criterion, the parameter identifiers involved, and the responsible role.

Five requirements need verification procedures that do not yet exist anywhere:

- UMR-092 session workspace isolation and ephemeral destruction.
- UMR-093 code execution behaviour.
- ASG-011 code containment. This needs an adversarial procedure: a containment boundary is only proven by trying to break it.
- UMR-094 and ASG-012 retrieval tool containment. Also adversarial: the procedure must attempt free-form command injection through the parameterised interface, and must attempt to read a document the requesting user cannot open in the source system.

- UMR-004 retrieval quality across every path, not only the default pipeline. The acceptance set under UMR-064 must exercise agent-selected retrieval, or it will measure a path production rarely takes.

Annex E — Open points

OP	Question	Status	Blocks
OP-01	Parent document	Closed. Arch and Design document will not be used as a parent	Section 2.1
OP-02, OP-03, OP-09	Missing appendices	Closed. Will use as RD-04, RD-05, RD-07	Annex C
OP-04	Security profile and ceiling	Closed. PBMM, Protected B	UMR-058
OP-05, OP-18, OP-22	Hosting platform and region	Closed. Azure, Canadian regions	UMR-070
OP-06, OP-13	Identity and separation of duties	Closed. Azure identity, role list in Section 3.4	UMR-065
OP-07	Records retention schedule	TBC by CSA decision	UMR-073, PARAM-27
OP-08	Risk standard	TBC by CSA decision	UMR-036
OP-10	French-language quality standard	Closed. Not needed at this phase	UMR-046, UMR-075
OP-11	Algorithmic Impact Assessment	Closed by CSA decision.	UMR-071
OP-12	Fixed or configurable targets	Closed. Both stay fixed	UMR-056, UMR-059
OP-14	CRAFT does not claim compliance with the Directive on Automated Decision-Making, but holds Protected B data, reads the financial system of record, runs generated code under UMR-093, and now selects its own	Phase 1 UC. Recommend review at the PBMM authorization	UMR-071, UMR-093, UMR-094

OP	Question	Status	Blocks
	retrieval strategy under UMR-094. Confirm the position still holds, and name the owner		
OP-15	Report generation target	Closed. Runs on demand under UMR-090	UMR-056
OP-16	Ceiling conflict	Closed. Resolved by the Protected B ceiling	UMR-047, UMR-053
OP-17	Unlabelled documents treated as Unclassified	Decided, carried as a residual risk. Review before production authorization	UMR-058, UMR-067
OP-19	Named Backup approver	TBC by CSA decision	UMR-025, PARAM-37
OP-20	SharePoint migration timing	Closed by CSA decision. Deferred to phase 2	UMR-052
OP-21	Owner of the simulated-to-real verification gate	TBD by CSA decision	UMR-091
OP-23	Traceability coverage	Closed in P11	Annex C
OP-24	Use case identifier collision	Closed in P10	Annex C
OP-25	Appendix A conflict	Closed in P10, superseded by OP-27	Section 5.2
OP-26	Document number and approval block	Closed. CSA-CRAFT-RD-0001; "CIO?" removed	Section 2.1, 2.4
OP-27	On-premises figures versus Azure	Closed. Azure; on-premises sizing withdrawn	Section 5.2
OP-28	No costed infrastructure position exists until	5K\$ from SSC. Then MB to assume the	Section 5.2, ADR-18

OP	Question	Status	Blocks
	RD-01 derives Azure sizing. Confirm who owns the Azure costing exercise	cost	
OP-29	UC-C8 and UC-C9 were written by this review. Confirm both at the Requirements Review	Confirmed. Both UC were modified as per MB and JFC approval email.	Annex C
OP-30	Agent architecture: registered roles, on-the-fly roles, or something else	TBC	UMR-011, UMR-012
OP-31	Code execution needs PARAM-42 limits and an approved library list. Neither exists. Code execution cannot be enabled without both	Postponed for a future decision.	UMR-093, ASG-011
OP-32	Session workspace retention	Ephemeral by default.	UMR-092
OP-33	Who builds and owns the Phase 1 Engineering Costing corpus, how is it version-controlled, and is it varied enough to support the acceptance data set in UMR-064?	TBC	UMR-051, UMR-064
OP-34	The authorized retrieval tool set under UMR-094 must be defined and approved before agent-selected retrieval is enabled. PARAM-43 iteration limit is also unset	TBC	UMR-094, PARAM-43, ADR-22

OP	Question	Status	Blocks
OP-35	Session lifetime and idle timeout (PARAM-44) are unset. Without them, sessions, agent contexts and workspaces have no defined end, which conflicts with the ephemeral default in UMR-092	TBC	UMR-095, PARAM-44
OP-36	The delegation criteria in PARAM-45 must be defined before delegation is enabled, or “demonstrably improves quality, speed, isolation or capability” cannot be tested and delegation becomes discretionary in practice	TBC	UMR-011, AAR-010, PARAM-45

Annex F — Reconciliation of the technical review

Source: RD-08, requirements-analysis.docx, with the CRAFT analysis responses. Adopted means the requirement now says what the response says. Adopted with control means the capability is admitted and bounded. Clarified means the requirement meant this already and now says so. Reversed means said something the response rules out.

RC (Review Comments)	Point	Affects	Disposition
RC-01	Document sources. Phase 1 should use only a controlled Engineering Costing corpus. Personal devices, individual OneDrive accounts and production CSA repositories require	UMR-051, UMR-052, UMR-091, Section 1.3	Adopted. UMR-051 names the Phase 1 scope and the five onboarding controls. Connectors deferred. OP-33 asks who owns the corpus

RC (Review Comments)	Point	Affects	Disposition
	a separately approved onboarding process with ownership, permissions, classification, retention and audit controls		
RC-02	8,192 tokens is the minimum total request context, not the size of an embedding chunk	UMR-003, UMR-067, PARAM-02, PARAM-39	Clarified. UMR-003 states the scope of the figure. Chunk size separated into PARAM-39
RC-03	Larger-context models are permitted, but retrieval must remain selective and relevant rather than filling available context indiscriminately	UMR-003, UMR-010	Adopted. The selectivity rule is now in UMR-003. This matters for grounding: filling a long window with weakly relevant passages lowers answer quality and weakens the UMR-010 measure
RC-04	Do not blindly append the top results from each method. Normalise, fuse, deduplicate	UMR-004, ARR-002, PARAM-03	Adopted, and this reverses V2 . V2 recorded “top 5 of each and append” as an acceptable Phase 1 fallback. UMR-004 now requires normalisation, fusion and deduplication, and prohibits appending
RC-05	Validate thresholds such as 0.72 against measured retrieval performance	UMR-004, PARAM-03	Adopted, and this partly reverses V2 . P9 removed 0.72 outright. It returns in PARAM-03 as a

RC (Review Comments)	Point	Affects	Disposition
			candidate to validate, not as a fixed rule
RC-06	Hybrid retrieval through a common interface, initially 0.7 semantic and 0.3 lexical	UMR-005, ARR-002, PARAM-04	Adopted. Fixed Phase 1 weights behind a common interface
RC-07	Do not use multiplied token probabilities. They do not indicate correctness and are often unavailable from hosted models	UMR-007, ARR-004	Adopted. Explicitly prohibited
RC-08	Base confidence on evidence coverage, citations, traceability, source agreement, retrieval quality, and identification of unsupported claims	UMR-007, ARR-004, UMR-081	Adopted. All six inputs are named in UMR-007. Also removes a hidden constraint on UMR-081, which requires a replaceable model
RC-09	“Agent spawning and termination without interruption” implies persistent agents tied to a session	UMR-016, UMR-095	Clarified. UMR-016 concerns platform availability. Session persistence is UMR-095
RC-10	Provide a session-scoped primary agent with an isolated, ephemeral workspace by default. It may use authorized retrieval and analysis tools and generate artifacts, but all actions,	UMR-092, UMR-093, UMR-094, UMR-095, ASG-011, ASG-012, UMR-030, UMR-082	Adopted with control. The workspace default changes from retention-TBC to ephemeral. Export control added. Authorized-tools-only is now the governing rule for both retrieval and

RC (Review Comments)	Point	Affects	Disposition
	permissions, exports, resource use and retention must be controlled and auditable		code
RC-11	Episodic memory is governed conversation and task history	UMR-014, AAR-004	Adopted. Definition used verbatim
RC-12	Semantic memory is curated, source-backed durable knowledge	UMR-014, Section 1.8	Adopted. “Curated” and “source-backed” are now requirements: an entry must cite its source and be reviewable
RC-13	Procedural memory is approved guidance such as instruction files, runbooks and templates	UMR-014, PARAM-40	Adopted. “Approved” is the operative word: procedural memory is version-controlled and changed only through the approved process
RC-14	Do not permit uncontrolled self-modification of procedures	UMR-028, AAR-009, Section 1.3	Adopted. Prohibition extended from settings, prompts and code to procedural memory
RC-15	Do not impose fixed roles or use multiple agents by default	UMR-011, UMR-012, AAR-001	Adopted, and this reverses V2 . The registered-role-per-agent rule is withdrawn. Control moves to the tool boundary in UMR-012
RC-16	Use one accountable primary agent that delegates	UMR-011, AAR-010, PARAM-45, ADR-03	Adopted. OP-30 closed. UMR-011 rewritten. AAR-010

RC (Review Comments)	Point	Affects	Disposition
	bounded subtasks only where this demonstrably improves quality, speed, isolation or capability. It remains responsible for validating and integrating the final response		added for validation and accountability. PARAM-45 makes “demonstrably” measurable
RC-17	Gap 1. The agent should choose how to search, including dense retrieval, sparse retrieval, or constructing commands such as grep or find	UMR-094, ARR-010, ASG-012, UMR-066, UMR-064, UMR-080	Adopted with control. The agent selects among authorized retrieval tools and may iterate. File search is parameterised. Free-form shell strings are prohibited, because the command space is unbounded, unauditable and open to injection from ingested content. Permission trimming applies to every path
RC-18	Gap 2. Expose a chat interface with a persistent agent to the user	UMR-095, UMR-001, UMR-014, UMR-092	Adopted. One primary agent per session, persisting for the session, holding episodic memory, bound to the workspace. Session, agent context and workspace end together
RC-19	Gap 3. The response text embedded in	Annex F	Closed. The responses were

RC (Review Comments)	Point	Affects	Disposition
	RD-08 is truncated mid-sentence		supplied separately and are reconciled above as RC-01 to RC-18. The reconciliation is now complete against both the comments and the responses